

pler than DC motor starters.

- The induction motor creates very low noise, vibration, etc.

Their disadvantages:

- Throughout the light-load situation, the power factor is extremely less, and it draws a huge current. So, the copper loss can be high, which decreases the efficiency throughout the light load situation.

- The squirrel cage induction motor's initial torque is not low.

- This is an invariable-speed motor, so cannot be used where variable speed is required.

- This motor's speed control is not easy.

- The motor includes a high starting inrush current, which causes a reduction in voltage at start time.

Their applications:

- Lifts
- Cranes
- Hoists
- Large-capacity exhaust fans
- Driving lathe machines
- Crushers
- Oil extracting mills
- Textile industry, etc

Permanent magnet motors

Permanent magnet motors generate torque by interacting stator currents with permanent magnets on or inside the rotor. Surface rotor magnets are common in small- and low-power motors for commercial machines, IT equipment, and automotive auxiliaries. Internal magnets are trivial on larger machines like electric vehicles and in industrial motors.

In permanent magnet motors, stator can use concentrated (short-pitch) windings if or when torque ripple is not a problem, but distributed windings are the criteria in larger motors. Since these motors do not have mechanical switches, inverters are essential to control their winding current.

Unlike other types of brushless motors, permanent magnet motors do not require current to sustain their magnetic fields. Therefore, the motors provide maximum torque per volume and may be the best choice if light

weight or small size is important. The absence of magnetising current also means greater efficiency in the 'sweet spot' load, that is, where the motors perform the best.

The most visible disadvantage of PMM is their high cost. High-performance permanent magnet motors use magnets made from transition metals such as dysprosium and neodymium. These materials are rare and come from geopolitically unstable countries, resulting in high and volatile prices.

In addition, while permanent magnets provide performance gains at lower speeds, they are also a kind of technical Achilles' heel. For example, as the speed of a motor increases, its rear EMF approaches the inverter supply voltage, making it impossible to control the winding current. This defines the basic speed of a generic permanent magnet motor, and in surface magnet designs it generally represents the maximum possible speed for a given supply voltage.

At speeds higher than the base speed, the motors utilise weakening of the active field in which the stator current is varied to deliberately depress the flow of the magnet. The speed range in which this can be successfully done is limited to about 4:1. As before, this limitation can be circumvented by reducing the winding curves and accepting higher costs and energy losses in the inverter.

The requirement to weaken the field is related to speed and the associated losses arise heedless of the torque. This lowers efficiency at high speeds, notably for light loads. This happens in electric vehicles, where cruising at highway speeds inevitably

involves weakening the field. Permanent magnet motors are often favoured for EVs, but the efficiency gain is questionable when computed in real-world driving cycles. In an interesting observation, at least one major EV manufacturer has switched from permanent-magnet to induction motors.

Another disadvantage of the motors lies in the fact that they are infamously challenging to handle in failure conditions due to their built-in rear EMF. Current continues to

flow in winding faults as the motor spins, even if the inverter is not connected, causing overheating and cogging torque, both of which can be dangerous. The loss or weakening of the field at high speeds, for example due to shutdown of the inverter, leads to uncontrolled generation and the voltage of the inverter DC bus can increase to dangerous levels.

One more significant limitation is the operating temperatures for all permanent magnet motors, except those equipped with samarium-cobalt magnets. And high motor currents, which can arise due to inverter failures, can force demagnetisation.

The maximum speed for the motors is defined by the mechanical retention of the magnet. If a motor is damaged, repairing it normally involves sending it back to the factory due to the difficulty of safely pulling out and handling the rotor.

Finally, recycling at the end of its useful life is also problematic, although the high current value of rare earth materials can make this more economically viable.

Despite these flaws, these motors remain unrivalled in terms of low speed and sweet spot efficiency and are useful when weight and size are critical.

Permanent magnet synchronous motors

The main difference between a permanent magnet synchronous motor and an induction motor lies in their rotors. Studies show that a synchronous motor has an efficiency of approximately 2% more than a highly efficient (IE3) induction electric

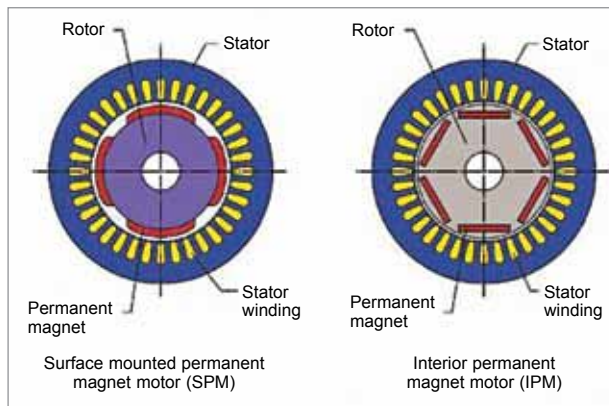


Fig. 3: Cross-sections of permanent magnet motors